

	Case Name: Social Apartments Ypres (3)	Sector	Construction (residential building)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
Submitted by	Matilde Casado and Margarida Antunes		One of nine std. scenarios
Date	2015		User defined distributions
File Name	C2015-32 Social Apartments Ypres (3)	Project Control	Automatic tracking
			Tracking based on user input

1. Project description

Project authenticity

The project involves foundation work, construction, utilities installation, roofing, and finishing.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	53	Serial/Parallel (SP)	38%
Planned Duration (PD)	358 days*	Activity Distribution (AD)	63%
Budget At Completion (BAC)	2 509 031.4 €	Length of Arcs (LA)	3%
Renewable Resources	-	Topological Float (TF)	59%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity				Time sensitivity		
	avg [%]	std dev [%]	skew [-]		avg [%]	std dev [%]	skew [-]
CRI-r	11.9	12.5	-3.5	CI	40.2	46.7	-0.4
CRI-rho	14.6	13.9	-2.6	SI	51.6	44.1	0.0
CRI-tau	25.5	23.8	-0.8	SSI	10.1	13.7	-1.4
				CRI-r	12.3	12.3	-1.7
				CRI-rho	13.2	13.0	-1.7
				CRI-tau	10.2	15.1	-4.4

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	19.6	-9.3	1	4.5	0.7
PV - SPI	40.5	37.3	CPI	6.7	-0.2
PV - SCI	41.3	36.1	SPI	34.5	34.5
ED - 1	1387.1	1378.7	SPI(t)	22.7	22.7
ED - SPI	40.5	37.3	SCI	34.1	34.1
ED - SCI	40.4	36.8	SCI(t)	22.7	22.1
ES - 1	13.9	-13.9	0.8 CPI + 0.2 SPI	22.9	22.2
ES - SPI(t)	15.8	10.9	0.8 CPI + 0.2 SPI(t)	9.2	8.0
ES - SCI(t)	16.0	10.2			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Automatic tracking was performed over 14 tracking periods with a length of 1 month. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on simulation results.

Authenticity assessment is not relevant here as it is not possible to introduce any kind of tracking information obtained from the actual project owner when performing automatic tracking. Activity durations and corresponding costs were generated based on the distribution profiles described in section “2.2. Risk Analysis”.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-27667.0	338470.4	265.9	1.0	1.6	1.2	1.0
std dev	51566.0	383194.7	215.5	0.1	0.7	0.1	0.1
final	0.3	407120.3	744.0	1.0	1.2	1.4	1.0

2.3.3.2. Time forecasting

PD	358 days
----	----------

Real Duration	266 days
---------------	----------

Early	-25.7%
-------	--------

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	412.9	3.0	2.2	-2.2
PV - SPI	413.6	3.2	2.2	-2.2
PV - SCI	414.4	3.2	2.2	-2.2
ED - 1	415.1	3.0	2.2	-2.2
ED - SPI	415.8	2.9	2.3	-2.3
ED - SCI	416.5	2.9	2.3	-2.3
ES - 1	417.2	2.9	2.3	-2.3
ES - SPI(t)	418.0	3.0	2.3	-2.3
ES - SCI(t)	418.6	3.2	2.3	-2.3

2.3.3.3. Cost forecasting

BAC	2 509 031.4 €
-----	---------------

Real Cost	2 509 031.5€
-----------	--------------

On Budget	0%
-----------	----

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	2536698.7	51566.0	0.1	-0.1
CPI	2544417.1	266603.0	0.1	-0.1
SPI	2115819.5	422255.9	1.1	1.1
SPI(t)	2326290.0	160532.7	0.5	0.5
SCI	2143681.1	500866.7	1.0	1.0
SCI(t)	2334671.0	309197.8	0.5	0.5
0.8 CPI + 0.2 SPI	2411884.0	297633.4	0.3	0.3
0.8 CPI + 0.2 SPI(t)	2489328.9	225313.3	0.1	0.1